

RESEARCH ARTICLE

An Economic Framework for Space Immigration

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Large-scale, voluntary space settlement must be economically rational to be viable. Here, we deploy the Roy model, an economic model used to understand immigration, to illuminate the economic factors important for space settlement and develop qualitative understanding of robust features of space settlement that do not depend on the details of the space economy. We find that getting the cost of living in space down by approximately 2 orders of magnitude is necessary to generate a space population on the order of 1 million people and the typical net utility of immigrating will be on the order of this cost. In addition, if the space economy is driven by productive activities of space settlers and there is some correlation between Earth and space skills and income, space settlers are likely to be drawn from the upper tail of Earth income distribution. An ideal way to incentivize immigration by these high-skill, high-income individuals is to declare the space economy free of redistributive taxes. Alternatively, if space settlement is driven by an insurance policy on civilization involving monetary transfers from Earth to space settlers, the space settlers are likely to be drawn from the lower tail of Earth income distribution, and only minimal marginal income beyond the cost of living in space will be necessary to create positive net utility of immigrating for them. The usefulness of the Roy model is demonstrated by its flexibility in providing qualitative insight in these disparate situations.

Introduction

Plans are currently being made for self-sufficient space settlements with populations on the order of 10^6 people on the Moon, Mars, large space stations, and other destinations [1–3]. Although a few thousand enthusiasts might wish to settle space for a mix of reasons, a settlement of 1 million people likely requires a viable economic model. Moreover, we will assume here that space settlement occurs in the context of a free society, rather than a command economy, so that space settlers must freely choose to immigrate on the basis of some perceived economic or other benefit. This dovetails with the transition in actors driving space exploration from government agencies to private companies [4]. Although there has been some discussion of potential economic activity in space [1,2,5], no economic modeling of the factors that will control the population of individuals who choose to immigrate to space has been performed. The purpose of this paper is to apply the Roy model to this problem.

The Roy model is the standard workhorse for studying the economics of immigration on Earth. It is derived from the work of Roy [6], who was interested in why workers choose between 2 different industries in the same location as well as the characteristics of workers who end up in each industry. Borjas [7] extended the Roy model to consider immigration by placing the 2 industries in separate locations and including a cost of immigration. We describe the Roy model and our application of it in detail in Materials and Methods. Briefly, the model provides a framework for understanding why individuals might choose to immigrate. Each individual has skills that determine

a potential income in the new location and the location of origin. The choice to immigrate is driven by the economic net utility of the decision, or the net income in the new location less the cost of immigration, compared to the net income in the location of origin. Individual potential net incomes are drawn from distributions relevant for the 2 locations. Our application of the Roy model differs from typical Earth-based applications primarily in the fact that the cost of immigration to space is very high, and as a result, the fraction of the population that chooses to immigrate will be low.

Our work can be applied to space settlement generally. That said, most work in this area has been focused on the settlement of Mars, and we can use the Mars case to get some preliminary estimates of economic variables. A variety of models have been proposed for Martian economic development. For example, Zubrin [2] suggests inventions and intellectual property, real estate, tourism, luxury goods, spectator sports, material exports, and a triangle trade with Mars as a base for asteroid mining. Neukart [5] identifies a list with some overlap with Zubrin including extraction and utilization of Martian resources, technological development and intellectual property, tourism and real estate development, collaborations and public relations (companies pursue Mars settlement to generate interest and business), research and development tax incentives, and supply chain and market expansion. Moreover, local economic activity could involve in situ resource utilization (ISRU) to generate oxygen [8–10], water [11,12], planetary warmth [13,14], fuel [15–18], and construction material [19–21], among other things. In this paper, we will not specify a specific income-generation model in space due to current uncertainty and, instead, explore the

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qualitative predictions and understanding we can gain about space settlement using the Roy model framework.

In addition to income generation in a space settlement, the Roy model requires an estimate of the annual cost of surviving in space relative to Earth, including the amortized cost of transport. Space settlers might move to space permanently, in which case the transportation cost would be paid only once, or they might do a tour in space to generate wealth that would be saved and used later for consumption on Earth, in which case the transportation cost would be paid twice. For our purposes, we need only an order of magnitude estimate of the amortized transportation cost, so it will be sufficient to divide the one-way transportation cost by 10 years, which will likely be the order of magnitude of time spent in space, whether the stay is temporary or permanent. Again, the case of Mars has been worked out in most detail, so we can draw from it for order-of-magnitude estimates. The current cost of transportation to Mars is about \$300 million per person [2], but reusable rockets could bring this down to \$300,000 [2] or even \$30,000 [1]. This translates into a current amortized cost of \$30 million per year currently that could be reduced to \$3,000 to \$30,000 per year on a multidecadal timescale. Currently, the cost of keeping a person alive on Mars is in the range \$10 million per year [1], so the current per year cost of moving to Mars would be about \$40 million and is dominated by transportation. The cost of surviving on Mars will clearly decrease because of economies of scale, as a settlement of 1 million people is built up, but it is difficult to estimate definitively how much. Moreover, the necessities of survival will drive innovation in ISRU, further decreasing costs, so ISRU can both increase space income and decrease the cost of surviving in space. Below, we will consider the case where the cost of surviving in space is reduced to \$100,000 per year, 2 orders of magnitude lower than the current cost of surviving on Mars, and the future amortized cost of transport is negligible relative to this.

In what follows, we will fully specify the Roy model for space settlement in Materials and Methods and apply it in Results. In doing so, we will focus on identifying the major qualitative drivers of space settlement from an economic standpoint. This will allow us to make some robust predictions about space settlement and the characteristics of space settlers, and it will also help focus future work to optimally constrain parameters and generate more quantitative predictions. Finally, we will discuss lessons learned and prospects for further development of these ideas in Discussion.

Materials and Methods

Definition of model

In the Roy [6] model, individuals sort themselves into the occupation or location that offers them the highest utility. Utility in the new location is defined as potential net income in the new location (here, space) minus the cost of immigration, and utility in the location of origin is defined as potential net income in the location of origin (here, Earth). The model considers the real income in 2010 US dollars, after subtracting taxes and adding (or subtracting) the monetary value of location-specific amenities. Amenities are features such as schools, parks, natural beauty, and safety that make a location more or less attractive. In the context of space settlement, a sense of adventure or a sense of the intrinsic value of human exploration of space could also be included in the amenities term. Following Roy [6], we assume

that each individual's potential, real income on Earth (w_0) and in space (w_1) is drawn from a log-normal distribution,

$$w_{0i} \sim 10^{\mathcal{N}(\mu_0, \sigma_0^2)} \quad (1)$$

$$w_{1i} \sim 10^{\mathcal{N}(\mu_1, \sigma_1^2)} \quad (2)$$

where i indexes individuals, μ_0 (Earth income scale) is the mean of the log base 10 of Earth income in dollars, σ_0 (Earth income inequality) is the standard deviation of the log base 10 of Earth income in dollars, μ_1 (the space income scale) is the mean of the log base 10 of the space income in dollars, and σ_1 (the space income inequality) is the standard deviation of the log base 10 of the space income in dollars. We further assume a covariance matrix between the draws of Earth and space income defined by a correlation coefficient between the 2 incomes of ρ .

Our accounting model for real, net income of individual i in location j is given by

$$w_{ji} = p_0 \frac{m_{ji} - t_{ji} + a_{ji}^g + a_{ji}^n}{p_j} \quad (3)$$

where m_{ji} is the nominal income, t_{ji} is the nominal tax required to cover the cost of government-provided amenities (excluding those that support survival), a_{ji}^g is the nominal monetary value of such government-provided amenities, a_{ji}^n is the nominal monetary value of natural amenities in a location, p_j is the local price level, and the factor p_0 converts income into Earth dollars. The purpose of this accounting is to show how we account for different material aspects of the conditions on Earth and in space and how they relate to the distribution of w_{ji} across individuals.

Unpacking this accounting model helps us understand what drives differences in real, net income across individuals. An individual would have a good sense of their Earth income potential. They would also have some sense of their nominal space income: Job offers from companies in space would account for their skill and convey their income in space. Focusing on after-tax incomes allows us to consider the effect of redistributive taxation on immigration by varying σ_0 and σ_1 . We exclude the cost of survival from our accounting model for real net, income in space for 2 reasons. First, it likely does not vary across individuals and the accounting model focuses on items that may vary across individuals (and, so, are indexed by i). Second, we account for cost of survival below, when we factor the incremental cost of living in space into the decision whether to live in space. We omit taxes that defray the cost of living from t_{ji} because that is included in costs of living in space below. Government-provided amenities (exclusive of those that support survival) include things such as income support (welfare), education, and public transportation. Natural amenities include things such as the view of a national park, the novelty of living on Mars, and the claustrophobia of living on a space station. Using real, net income, calculated by dividing the numerator by the price level, allows us to account for the impact of higher prices for consumption (other than survival) in space. The price level is calculated as the local price of a fixed and commonly consumed bundle of goods. The factor p_0 is the price of that bundle at a particular, arbitrary location, e.g., the United States in the year 2010. This converts the units of w_{ji} into 2025 US dollars. In this paper, we will focus on the composite variable w_{ji} rather than the individual effects of

$(m_{ji}, t_{ji}, a_{ji}^g, a_{ji}^n, p_{ji}, p_0)$. These individual effects are important, however, and should be studied in more detail in future work.

The second important input into individual utility in space is the cost of living in space, including the amortized cost of transport to space, C . We assume that this cost is constant across individuals because we do not see transport costs as an important source of interpersonal variation that explains why some people would go to space and others would not. If an individual is planning to return to Earth to consume after a period of higher wages in space, the transport cost must be paid twice. However, the price of consumption is higher for an individual making a one-way trip than a person making a return trip. The former must do all her consumption in space, where prices are very high, while the latter can save money and consume upon her return to Earth, where prices are lower. We do not explore this issue further in this paper, but, again, it would be an interesting focus of future work.

Following Borjas [7], the wage and living/transport cost terms can be combined to formulate the net utility U_i of immigrating to space for each individual

$$U_i = (w_{1i} - C) - w_{0i} \quad (4)$$

An individual is assumed to immigrate to space if $U_i \geq 0$ and remain on Earth if $U_i < 0$. Analytical approximations that work well for immigration between countries on Earth can be derived for the fraction of a population that has positive net utility associated with immigrating on Earth [7]. These approximations fail by many orders of magnitude in the high-cost, low-immigration context of space travel that we consider here. We therefore implement the model via Monte Carlo draws from large populations of hypothetical individuals, each with Earth and space potential incomes drawn from Eqs. (1) and (2). Using this method, we calculate the fraction of the total Earth–space population that will become space settlers. To find the space population, we multiply this fraction by 10 billion, the likely order of magnitude of the human population on a centennial timescale [8].

Using a total Earth-based population of 10 billion, a target space population of 1 million requires a space settler fraction of 10^4 . This fraction is small, so large numbers of draws will be necessary to achieve stable statistics. If a Monte Carlo algorithm involves N draws and yields N_e instances of positive net utility, we can estimate the true fraction of space settlers as $\hat{f} = \frac{N_e}{N}$ and

the 1σ uncertainty in this estimate as $\hat{\sigma}_f = \sqrt{\frac{\hat{f}(1-\hat{f})}{N}} \approx \sqrt{\frac{\hat{f}}{N}}$.

A stable estimate of the fraction requires $\frac{\hat{\sigma}_f}{\hat{f}} \approx \sqrt{\frac{1}{fN}}$ to be small or as a practical guideline roughly $N \approx \frac{100}{\hat{f}}$. For most simulations, we present that we draw 10^8 individuals, which means that we should be able to accurately estimate the space settler fraction down to about $\hat{f} \approx 10^{-6}$, corresponding to a space population of 10^4 . For some simulations, we draw only 10^7 individuals if this is sufficient for stable statistics.

Parameter values considered

We give the fiducial values of model parameters in Table 1. In Results, we will vary these parameters one at a time, while leaving other model parameters set to their fiducial values, to understand their relative importance in determining the space population.

We estimate the mean income (μ_0) on Earth and the variance of that income (σ_0)—a measure of inequality—by fitting a log-normal distribution to Earth's world income distribution in 2010 from Hong et al. [9] (Fig. 1). The data in Hong et al. [9] are from income tax returns and may underestimate the high-income tail of the distribution, but they are sufficient for the qualitative investigation here. Moreover, the log-normal fit overestimates the upper tail of the Hong et al. [9] income distribution, which may produce an offsetting error.

Given that we do not have a full specification of the future space economy, the space income distribution is highly uncertain. For the default case, we will simply assume that the space income distribution is identical to Earth income distribution ($\mu_1 = \mu_0$ and $\sigma_1 = \sigma_0$), and we will relax this assumption later to study the effect of changes in the space income distribution. We assume a fiducial correlation (ρ) of 0.9 between Earth and space incomes. This is reasonable on the basis of data from the Bureau of Labor Statistics' Occupational Employment Statistics Survey (www.bls.gov/oes), which suggests that the average correlation of wages across industries, conditional on occupation, is 0.94 (authors' data analysis). Finally, we assume an annual cost of living in space relative to Earth, including the amortized cost of transport (C) of \$100,000, as discussed in Introduction.

Results

We begin our analysis with a schematic plot of individual Earth and space incomes to illustrate basic features of the Roy model (Fig. 2). We can draw a number of intuitive lessons from this plot that we will investigate further below. First, the cost of living in space is a critical factor controlling the number of space settlers. At a minimum, the space income must be higher than this cost to make the net utility of immigration positive, so the cost has to get low enough that some of the space income distribution exceeds it to generate any space settlers. Beyond this, the lower the cost gets, the more space settlers there will be. Second, an increase in the real, net income in space relative to

Table 1. Fiducial values of model parameters

Name	Symbol	Value
Earth income scale	μ_0	3.64
Earth income inequality	σ_0	0.60
Space income scale	μ_1	3.64
Space income inequality	σ_1	0.60
Income correlation	ρ	0.9
Cost	C	\$100,000

Unless otherwise noted, parameters are set to their fiducial values in the simulations presented. Earth income scale (μ_0) is the mean of the log base 10 of Earth income in dollars. Earth income inequality (σ_0) is the standard deviation of the log base 10 of Earth income in dollars. Space income scale (μ_1) is the mean of the log base 10 of the space income in dollars. Space income inequality (σ_1) is the standard deviation of the log base 10 of the space income in dollars. Income correlation (ρ) is the correlation coefficient between Earth income and space income. Cost (C) is the annual cost of living in space relative to Earth, including the amortized cost of transport.

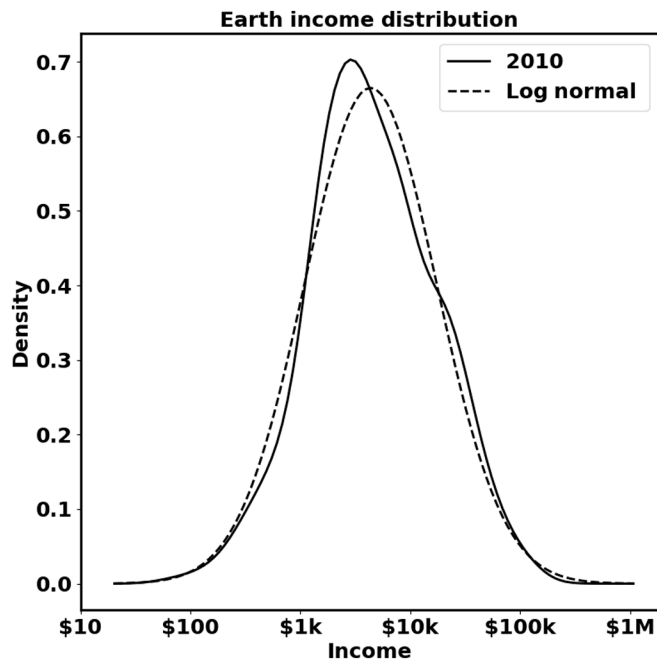


Fig. 1. Fit of a log-normal distribution to Earth's world income distribution in 2010 from Hong et al. [9]. The fitted parameters are a mean of 3.64 and standard deviation of 0.60 of the logarithm (base 10) of income in 2010 dollars.

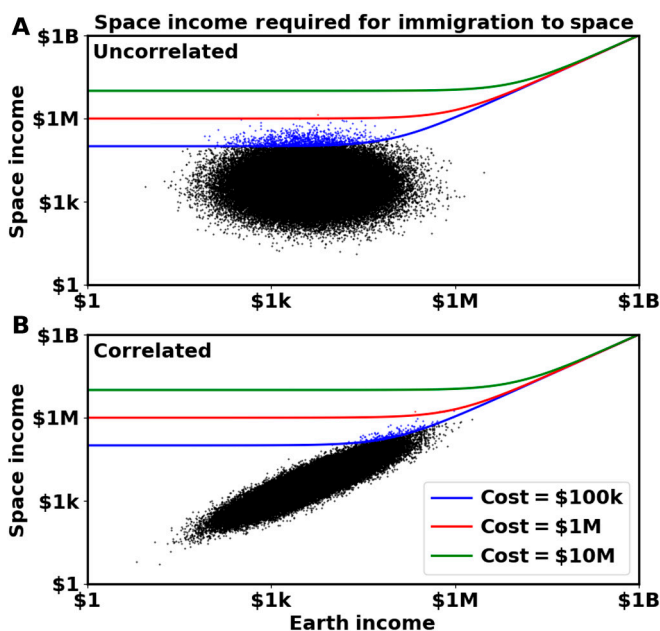


Fig. 2. Scatter plot of Earth income versus space income for 10^3 randomly chosen example individuals. Incomes for both Earth and space are drawn from the log-normal distribution approximating Earth's income distribution in 2010 (Fig. 1). In (A), Earth and space incomes are uncorrelated, and in (B), they have a correlation coefficient of 0.9. The lines represent the annual cost of living in space relative to Earth (C), including the amortized cost of transport, with the following values: \$100,000 (blue), \$1 million (red), and \$10 million (green). Individuals with a positive net utility (space income greater than the sum of Earth income and cost) are plotted with colors corresponding to the cost the income difference exceeds.

Earth increases the number of space settlers. This implies that the marginal productivity of labor in the space economy (a key determinant of nominal space wages), the risk of living in space

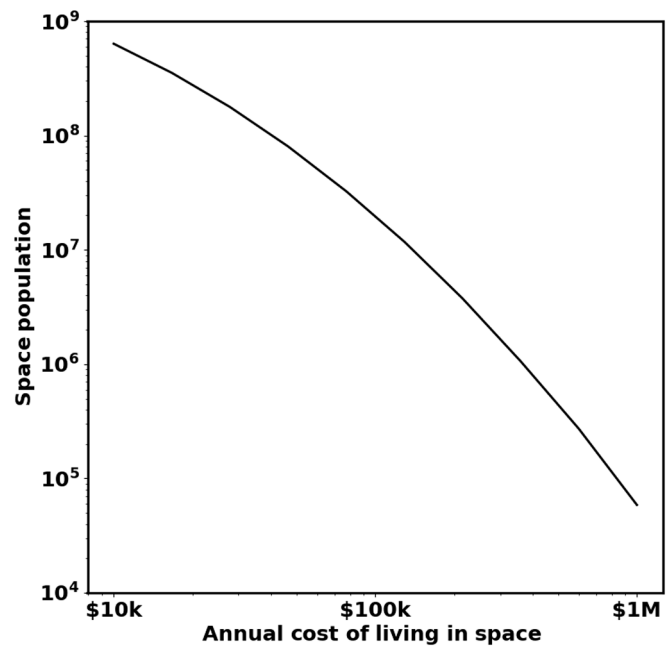


Fig. 3. Total population of space settlers as a function of the annual cost of living in space relative to Earth (C), including the amortized cost of transport.

(a determinant of the cost of living in and amenity value of space), and the price of goods in space (which determines the purchasing power of nominal space income) are critical determinants of whether a million-person space settlement is feasible. Third, an increase in Earth–space income correlation makes it so individuals with a high Earth income are much more likely to generate sufficient space income to offset costs than those with low Earth incomes. This implies that the space settlers will be drawn from the upper tail of Earth income distribution if there is a high Earth–space income correlation. Fourth, as a consequence of the previous point, increasing Earth–space income correlation decreases the number of potential space settlers and therefore the space population.

Cost of transport to and living in space

First, we examine the effect of the annual cost of living in space relative to Earth, including the amortized cost of transport, in more detail. As discussed in Introduction, this cost is currently in the range of \$40 million per year for hypothetical Mars settlers. At this cost, large-scale space settlement is completely infeasible for fiducial model parameters (Fig. 3). For fiducial model parameters, the cost must be brought down by about 2 orders of magnitude to around \$300,000 to achieve a space settlement with a population on the order of 10^6 . Reusable rockets and scalability due to mass production could get the price of a one-way ticket to Mars down to the range of \$300,000 [2] for an amortized cost of about \$30,000 per year. If this is achieved, then transport will actually represent a small component of cost of living in space. It will be critical to get the cost per year of surviving in a space environment down by approximately 1.5 orders of magnitude from \$10 million per year to the range of \$300,000 per year. Fortunately, as the size of the space settlement grows, the cost of surviving there should go down simply because of economies of scale and innovations in ISRU. In any case, our work points to studying this issue as a critical feature that strongly influences whether large-scale space settlement is economically feasible.

Relative income in space and income inequality

For our fiducial model parameter choices (Table 1), we assumed identical adjusted potential income distributions on Earth and in space due to lack of available information about how the space economy will function. Here, we will relax this assumption to examine the effect on space settlement of changing these income distributions. We can guide this discussion using the concepts of space pull and Earth push to drive space settlement. Space pull means that economic factors develop that make space more attractive, pulling immigrants to space. Earth push means that economic factors develop that make Earth less attractive, incentivizing immigration to space in search of better opportunities.

The most important income parameter is the scale of space income (Fig. 4A): A one order of magnitude increase in the space income scale leads to a roughly 4 orders of magnitude increase in the space population. This reflects the fundamental importance of developing and maintaining a viable space economic model for driving large-scale space settlement. Increasing the space income scale is a highly leveraged example of space pull. The scale of Earth income has an opposing effect (Fig. 4B). Earth push can drive space settlement if the scale of Earth income decreases, but the effect saturates in a way that space pull does not. On the other hand, increases in the scale of Earth income by about an order of magnitude (without commensurate increases in the scale of space income) can preclude large-scale space settlement. Assuming that the scale of Earth income continues to grow in the future, as it has in the past, this means that the scale of space income needs to keep pace or outpace this growth for large-scale space settlement to be economically viable.

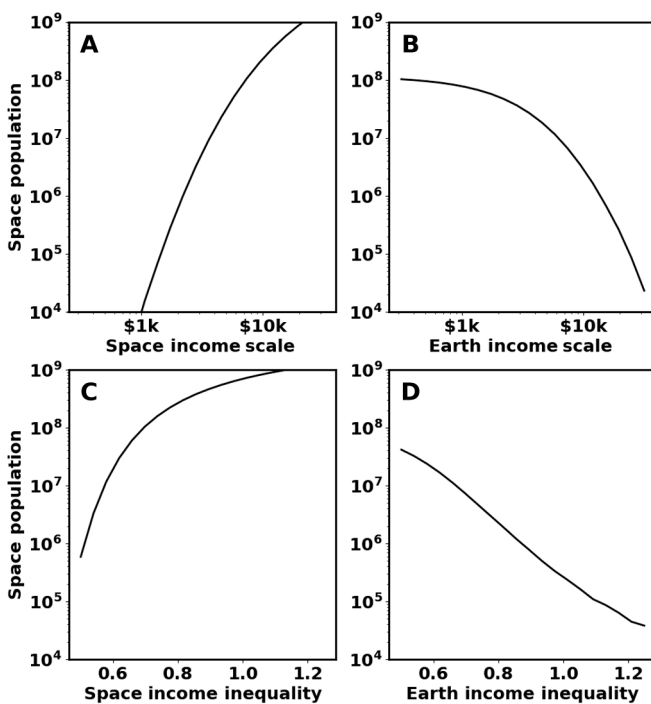


Fig. 4. Total population of space settlers as a function of (A) space income scale (μ_1 , mean of the log base 10 of the space income in dollars), (B) Earth income scale (μ_0 , mean of the log base 10 of Earth income in dollars), (C) space income inequality (σ_1 , standard deviation of the log base 10 of the space income in dollars), and (D) Earth income inequality (σ_0 , standard deviation of the log base 10 of Earth income in dollars). All parameters refer to the underlying, pretax income distributions, not to distributions conditioned on either remaining on Earth or moving to space shown in Fig. 7.

Income inequality can also be an effective driver of either space pull or Earth push. The larger the inequality in the underlying distribution of potential adjusted space incomes, the larger the space population will be (Fig. 4C). This reflects the fact that more variation in space skills and space amenity valuation will increase the number of individuals who find a space niche with net utility associated with immigration to space. It is also notable that increases in Earth income inequality lead to large decreases in the space population (Fig. 4D). For example, doubling Earth income inequality decreases the space population by about 3 orders of magnitude. Decreasing Earth income inequality can therefore be seen as an example of Earth push.

As discussed above, space income scale and space inequality have opposing tendencies on space pull and the resulting space population. We can see this directly by plotting the combination of space income scale and inequality necessary to drive 1 million space settlers (Fig. 5). Interestingly, the space income scale necessary to populate a space settlement is approximately the negative exponential of the space income inequality. This implies that small increases in space income inequality can be traded off with large decreases in the space income scale to maintain the same space pull, underlying the effectiveness of space income inequality in driving space settlement. Finally, as noted above, reducing the annual cost of living in space makes a space settlement possible for many more combinations of space income scale and space inequality.

Correlation between Earth and space income

In Fig. 6, we consider the effect of Earth–space income correlation in more detail. Using fiducial model parameters, zero correlation results in roughly 10^8 space settlers. Increasing the correlation to 0.9 decreases this number by roughly an order of magnitude, and the correlation must be increased beyond 0.99 to get the space population below 10^6 , our threshold for a self-sufficient space settlement. While the correlation in income across sectors conditional on skill has a correlation of roughly

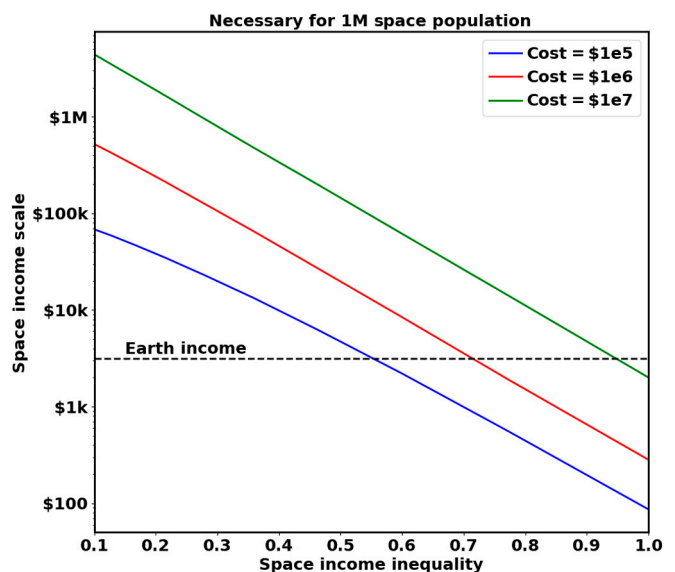


Fig. 5. Combination of space income inequality (σ_1 , standard deviation of the log base 10 of the space income in dollars) and space income scale (μ_1 , mean of the log base 10 of the space income in dollars) necessary to yield a total space settler population of 1 million for different annual costs of living in space relative to Earth, including the amortized cost of transport: \$100,000 (blue), \$1 million (red), and \$10 million (green).

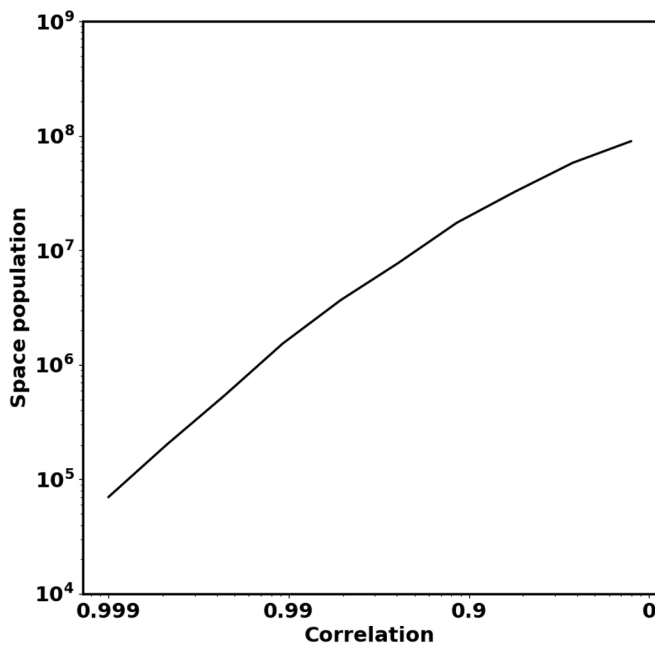


Fig. 6. Total population of space settlers as a function of the correlation coefficient between Earth and space incomes (ρ).

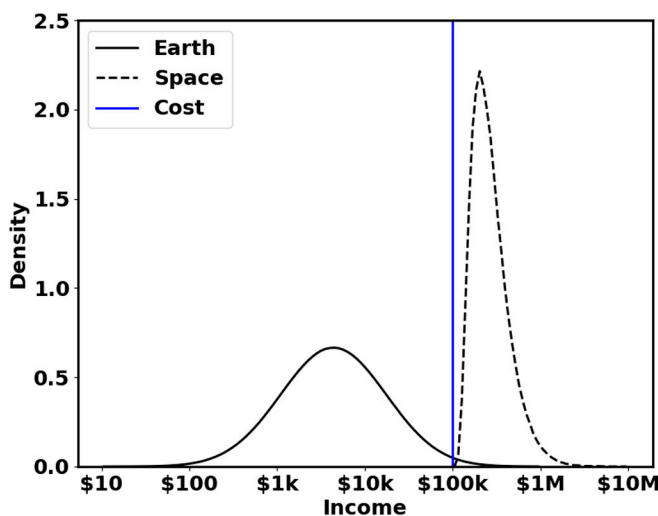


Fig. 7. Income distributions of those who remain on Earth (solid black) and of space settlers (dashed black) as well as annual cost of living in space relative to Earth (blue).

0.94 on average (see the “Parameter values considered” section), it is reasonable to expect differences in perceived amenities on Earth and space to keep the net income correlation well below that level. We can therefore conclude that although Earth–space income correlation reduces the space population somewhat and, as discussed above, influences the composition of the space population, it is unlikely to be a decisive factor in determining the feasibility of a self-sufficient space settlement. Experimentation with other parameter choices than those in Table 1 confirms this (not shown).

Space settler characteristics

As mentioned above, correlation between Earth and space incomes coupled with a large cost of living in space implies that space settlers will be drawn from the upper tail of Earth income

distribution (Fig. 2). We can see this more directly from the income distributions of those who remain on Earth and of space settlers in Fig. 7. Note that the income distributions plotted in Fig. 7 are conditional on either remaining on Earth or immigrating to space. The plot is made using fiducial model parameters, including identical adjusted potential income distributions for Earth and space. As expected, the minimum space income must exceed the cost of living in space, \$100,000 in this case, which corresponds to the top $\approx 1\%$ of Earth incomes for fiducial model parameters. Because of the log-normal nature of the income distributions, the mode net, real income of space settlers is about twice as large as the cost of living in space (Fig. 7), and the mode net utility of immigration of space settlers is on the same order as the cost of living in space (Fig. 8). This is an important additional prediction of the Roy model: Not only will space settlers be drawn from the upper tail of Earth income distribution with incomes generally exceeding the cost of living in space, but also their net utility of immigrating to space will be on the order of the cost of living in space. As costs are driven down, space immigration becomes attractive to more and more potential settlers, but the net utility of the move is also driven down correspondingly.

Discussion

In this paper, we have used the Roy economic model to understand the major qualitative factors that will affect the large-scale settlement of space with a self-sufficient population on the order of 1 million people. If space settlement is driven by productive economic activity, we can use the Roy model to derive the following main conclusions:

1. The cost of living in space, including transport, must be reduced by roughly 2 orders of magnitude relative to estimates based on current technology for about a million people to make enough money in space to offset this cost.
2. A viable business model that generates a scale of real, space income at least as high as that on Earth must be developed for large-scale space settlement to be feasible. We will discuss drivers of space income in more detail in this section.
3. Methods of inducing space settlement can be classified into the categories of “Earth push” and “space pull”, which we will discuss in more detail in this section.
4. The space settlers will be drawn from the upper tail of Earth income distribution. This is a robust prediction of the Roy model assuming some correlation between Earth and space income coupled with a high cost of moving to space.
5. The typical net utility of immigrating to space will be of the order of magnitude of the cost of living in space.

The Roy model is useful for understanding the supply of labor in space. The actual income earned in space, however, is a result of both the supply of labor and the demand for labor in space. In a free market, the equilibrium nominal wage is determined by the intersection of the supply and demand for labor in space. This paper has focused solely on the supply side of the labor market. The demand for labor in space, however, would be driven by the business model that generates revenue to pay nominal space income and the marginal product of labor in space.

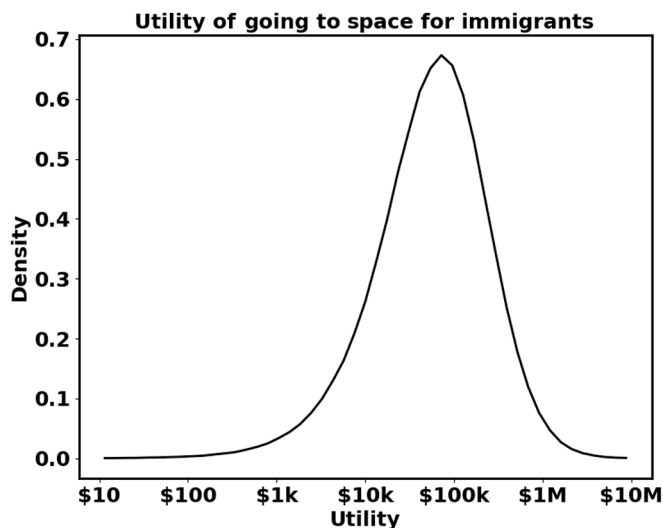


Fig. 8. Utility, measured in dollars, of moving to space for space settlers. Space settlers are those with space income larger than Earth income plus cost of living in space relative to Earth.

Valuable economic activities in space would be a source of labor demand, wages, and thus space pull. It must be admitted that current ideas for productive space activities [2,5] are vague and not entirely convincing. If we consider the proposal of mining, for example, important questions remain unspecified, such as exactly which minerals would be mined on asteroids or Mars, what prices could be expected for them if markets were flooded with extraterrestrial resources, what the cost of bringing these minerals to Earth would be, and why humans would do this rather than robots. Our work suggests that going through the exercise of careful economic studies of each proposed economic model to estimate the demand for human labor in space would go a long way toward increasing quantitative specificity of the viability of plans for space settlement.

There may also be important Earth push pressures that increase labor supply to space activities. If conditions on Earth worsen because of war, climate catastrophes, or malevolent artificial intelligence, and these reduce wages and utility on Earth, it would increase labor supply to space. We take no position on the likelihood of these scenarios. However, conditional on their real income on Earth, individuals workers who do judge these risks to be high are more likely to find space attractive.

Our income inequality results (see the “Relative income in space and income inequality” section) suggest a novel Earth push, space pull interaction mechanism for space settlement. To understand this, we need to remember that the income distributions we are discussing are after redistributive taxes. A large redistributive tax on Earth coupled with zero redistributive taxes in space would therefore decrease Earth inequality and increase space inequality. Both of these changes would increase the space population (Fig. 4). Even if the underlying economy is weaker in space than on Earth, large-scale settlement of space can be driven simply by increasing progressive taxes on Earth and/or declaring space a tax haven. This mechanism would be effective primarily because, as discussed above, space settlers will be drawn from the upper tail of Earth income distribution.

Musk [3] and others have spoken of large-scale space settlement as a backup of civilization, in case it is destroyed on Earth.

Without judging the likelihood of this scenario, we will show here how the Roy model can be used to make predictions about how much it would cost and who the likely space settlers would be. This idea is different from bad conditions on Earth driving labor to space: The insurance policy requires space settlement before conditions on Earth worsen and that settlement would continue even if conditions on Earth do not deteriorate. This space-as-a-hedge strategy could be developed into an economic model via a global insurance policy whose premiums are monetary transfers from those on Earth to space settlers. The payoff of this policy to those on Earth is similar to the payoff to a life insurance policy: It helps ensure that people you have social and genetic ties to are sustained after your death. Of course, there may be less private demand for a civilization insurance policy than there is for a life insurance policy, given that the social and genetic link between the average person on Earth and the average person in a space settlement will not not as strong as those between the average person on Earth and their children. However, there is a public good aspect to civilization insurance: Any such policy benefits all of humanity, not just the individuals who pay the premiums, i.e., it is nonexcludable. Moreover, the policy is nonrivalrous, since the benefits of the policy are not diminished by the number of people on Earth who benefit from it. These considerations suggest a role for governments to subsidize the income of space settlers as a subsidy for civilization insurance.

Interestingly, Earth–space income correlation would be undefined in the case of a civilization insurance policy, since the space income would just be a constant. The effect of this is that the desire to reduce the cost of insurance would cause space settlers to be preferentially drawn from the lower tail of Earth income distribution. This selection effect is the opposite of our previous conclusion about selection, which was based on space settlers generating their own wealth through productive activities (cf., [10]). Generating net utility for this space settlers under an insurance policy would primarily involve paying the cost of surviving in space, with only minimal additional income necessary to incentivize immigration. For example, with fiducial model parameters, a payment of only \$25 per year in real income is sufficient to motivate the lowest million individuals on Earth income distribution. Assuming that the cost of surviving in space can be brought down to \$100 thousand per year, this translates into an annual cost of \$100 billion or 0.1% of current gross domestic product. We note that even in this scenario, some local space economy would likely still develop around ISRU, entertainment, restaurants, and other inherently local activities.

Although many positive amenities associated with space settlement immediately jump into the minds of those of us excited by exploration and adventure, it is important to consider negative amenities as well. These would be costs for any business model of space settlement because these disamenities would have to be compensated by space workers’ income. First, there is the increased risk of death. This could be incorporated into the Roy model through the term C , the cost of living in space. The result would be a wage premium (higher space income), as is observed for risky jobs on Earth when worker skills are held constant [11–14]. The wage premium required to attract workers to space should be proportional to the difference in survival probability between space and Earth multiplied by the value of life. That value of life is, in turn, a function of Earth income and the value of amenities on Earth.

Second, there is likely to be a negative amenity associated with social isolation in space. Individuals working on offshore oil rigs do not travel with their families and friends, and the same might be true for space settlers. Bringing friends and family to a space settlement would add transport and living cost that would not necessarily be offset by wages, especially in the case of children. This loss of companionship would lower the amenity value of living in space. This too would have to be compensated by a higher nominal space income in space to attract workers [15,16]. Indeed, isolation is one of the reasons that the daily pay for long haul truck drivers is much greater than pay for regional or local truck drivers [17].

One aspect of living in space that is related to the lack of companionship is the potential skew in the sex ratio of space settlers. Risky jobs and jobs with longer shifts tend to have workers who are disproportionately male [18,19]. These conditions likely apply to work in space settlements so that the sex ratio of space settlers too may be biased toward males. Beyond the problem of isolation, a male-biased sex ratio may also lead to antisocial behavior, as males tend to be more violent than females, especially when few females are present [20,21]. This would also lead to a higher cost of living in space, since space settlements would need a larger police force or restrictions on freedom to keep the peace. These factors would have to be compensated by a higher nominal space income to attract workers.

A third challenge of living in space is the higher price of everyday consumption in space. Not only is the cost of survival (including food, water, and housing) higher, but the cost of other consumption (e.g., nonessential clothing, convenience appliances, and entertainment) is also higher. It is possible to ship these items from Earth, but that would entail space transport costs. If these items are produced in space, there would be fixed costs of production that would be spread over a far smaller population than on Earth. Moreover, it may require specialized technology with higher marginal costs. One way to reduce the cost of nonessential consumption is for workers to defer this consumption until they return to Earth. However, that would require the worker to travel back to Earth (as opposed to goods being transported to space). In other words, workers may temporarily migrate to space rather than permanently settle there. This means that the cost of transport trades off with the price of nonessential consumption and thus purchasing power in space. Another way to put this is that paying worker transport costs twice increases purchasing power of nominal income in space.

As noted in the introduction, we have focused on a settlement on Mars here, but our work could also be applied to settlements on the Moon, a large space station, or other destinations. One of the other destinations, particularly the Moon, might actually have a settlement first due to proximity, although it might not necessarily grow to a self-sustaining population of 1 million depending on economic viability. Such a settlement might allow some of the predictions we have made here to be preliminarily tested. In addition, since the cost of transportation to the Moon is currently much less than that to Mars, a settlement on the Moon would allow insight into a cost regime that might be relevant on Mars later.

The flexibility of the Roy model in describing these various situations argues in favor of its use to study the supply of labor for space settlement. At this point, the most important work is to further specify model parameters with detailed economic

analyses to set concrete goals and make concrete predictions. We hope that this paper serves as a guide for this work.

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Data Availability

The data consist of a Python notebook and world income distribution data file from Hong et al. [9]. They are available at <https://github.com/dorianabbot/Space-Settlement-Economics>.

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An Economic Framework for Space Immigration

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Large-scale, voluntary space settlement must be economically rational to be viable. Here, we deploy the Roy model, an economic model used to understand immigration, to illuminate the economic factors important for space settlement and develop qualitative understanding of robust features of space settlement that do not depend on the details of the space economy. We find that getting the cost of living in space down by approximately 2 orders of magnitude is necessary to generate a space population on the order of 1 million people and the typical net utility of immigrating will be on the order of this cost. In addition, if the space economy is driven by productive activities of space settlers and there is some correlation between Earth and space skills and income, space settlers are likely to be drawn from the upper tail of Earth income distribution. An ideal way to incentivize immigration by these high-skill, high-income individuals is to declare the space economy free of redistributive taxes. Alternatively, if space settlement is driven by an insurance policy on civilization involving monetary transfers from Earth to space settlers, the space settlers are likely to be drawn from the lower tail of Earth income distribution, and only minimal marginal income beyond the cost of living in space will be necessary to create positive net utility of immigrating for them. The usefulness of the Roy model is demonstrated by its flexibility in providing qualitative insight in these disparate situations.

Image

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